

All About Aggregation

Grouping, summarizing, pivoting, and windowing

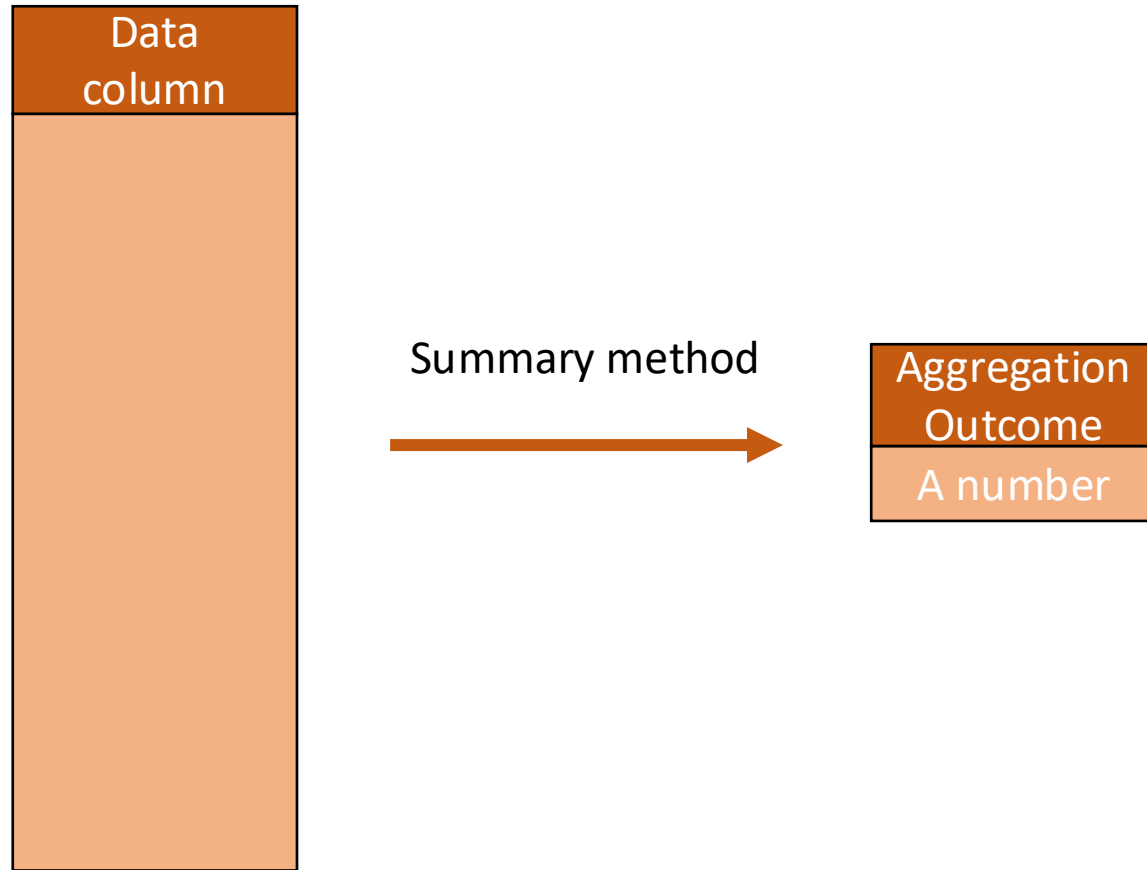
What is aggregation?

- Latin *aggregare* – “to flock together”
- In data terms, to *aggregate* is to create *one number* from *many numbers*
- The *one number* is a *summary*
 - Count
 - Mean
 - Median
 - Min
 - Max
 - Standard deviation

Simple aggregation

- *Simple aggregation* requires at a minimum 2 ingredients:
 - A data column
 - A summary method

Simple aggregation



Counting is simple aggregation!

	A	B	C	D
1	Station	Elevation	Month	Snowfall
2	BOULDER	5404	Jan	10.9
3	DENVER WSFO AP	5325	Jan	7.3
4	FOUNTAIN	5565	Jan	3.3
5	GUNNISON 1 N	7680	Jan	12.5
6	POWDERHORN	8094	Jan	7.2
7	BOULDER	5404	Feb	11
8	DENVER WSFO AP	5325	Feb	7
9	FOUNTAIN	5565	Feb	3
10	GUNNISON 1 N	7680	Feb	9.5
11	POWDERHORN	8094	Feb	6.9
12	BOULDER	5404	Mar	17.8
13	DENVER WSFO AP	5325	Mar	12.2
14	FOUNTAIN	5565	Mar	5
15	GUNNISON 1 N	7680	Mar	6.5
16	POWDERHORN	8094	Mar	5

NUMBER OF OBS

→ 15

Averaging is simple aggregation!

	A	B	C	D
1	Station	Elevation	Month	Snowfall
2	BOULDER	5404	Jan	10.9
3	DENVER WSFO AP	5325	Jan	7.3
4	FOUNTAIN	5565	Jan	3.3
5	GUNNISON 1 N	7680	Jan	12.5
6	POWDERHORN	8094	Jan	7.2
7	BOULDER	5404	Feb	11
8	DENVER WSFO AP	5325	Feb	7
9	FOUNTAIN	5565	Feb	3
10	GUNNISON 1 N	7680	Feb	9.5
11	POWDERHORN	8094	Feb	6.9
12	BOULDER	5404	Mar	17.8
13	DENVER WSFO AP	5325	Mar	12.2
14	FOUNTAIN	5565	Mar	5
15	GUNNISON 1 N	7680	Mar	6.5
16	POWDERHORN	8094	Mar	5

AVG
→ 8.34

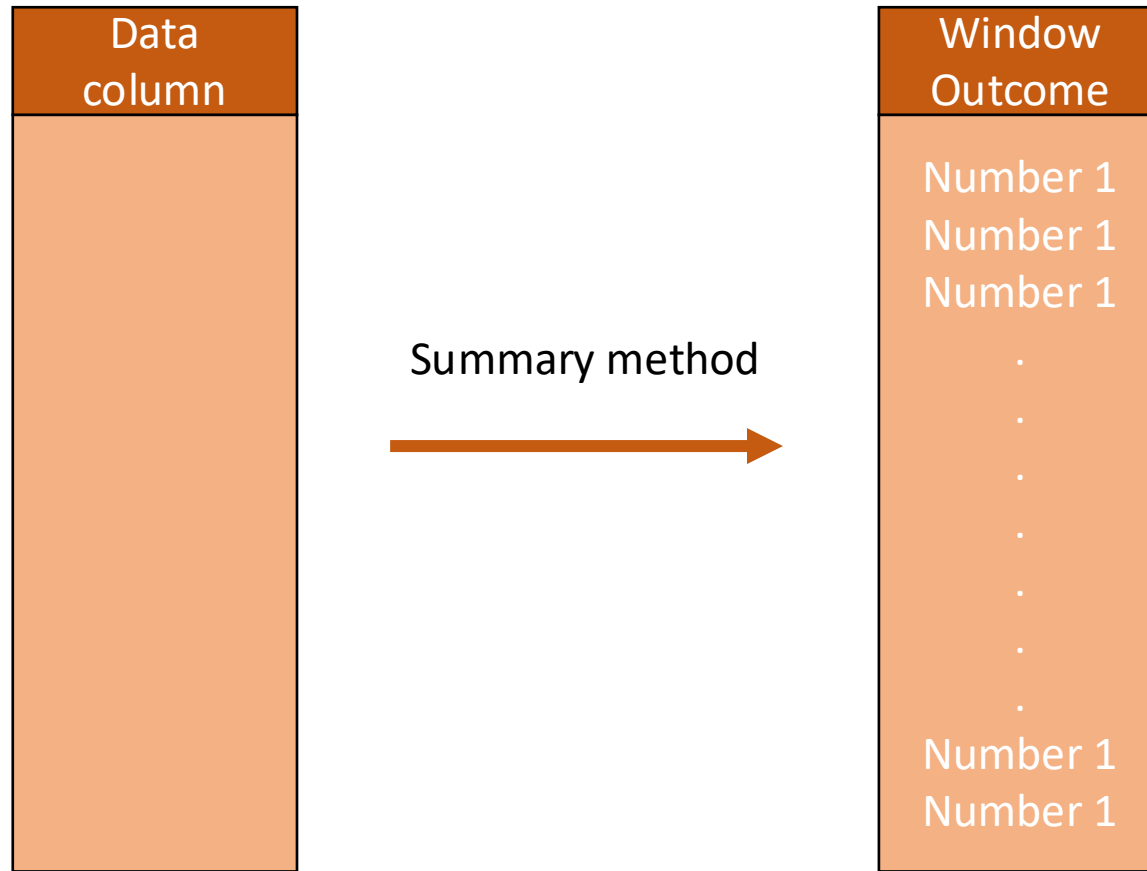
Maximum as a simple aggregation!

	A	B	C	D
1	Station	Elevation	Month	Snowfall
2	BOULDER	5404	Jan	10.9
3	DENVER WSFO AP	5325	Jan	7.3
4	FOUNTAIN	5565	Jan	3.3
5	GUNNISON 1 N	7680	Jan	12.5
6	POWDERHORN	8094	Jan	7.2
7	BOULDER	5404	Feb	11
8	DENVER WSFO AP	5325	Feb	7
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10	GUNNISON 1 N	7680	Feb	9.5
11	POWDERHORN	8094	Feb	6.9
12	BOULDER	5404	Mar	17.8
13	DENVER WSFO AP	5325	Mar	12.2
14	FOUNTAIN	5565	Mar	5
15	GUNNISON 1 N	7680	Mar	6.5
16	POWDERHORN	8094	Mar	5

MAX

8094

Simple Window Function



Averaging with a window function!

A	B	C	D
Station	Elevation	Month	Snowfall
BOULDER	5404	Jan	10.9
DENVER WSFO AP	5325	Jan	7.3
FOUNTAIN	5565	Jan	3.3
GUNNISON 1 N	7680	Jan	12.5
POWDERHORN	8094	Jan	7.2
BOULDER	5404	Feb	11
DENVER WSFO AP	5325	Feb	7
FOUNTAIN	5565	Feb	3
GUNNISON 1 N	7680	Feb	9.5
POWDERHORN	8094	Feb	6.9
BOULDER	5404	Mar	17.8
DENVER WSFO AP	5325	Mar	12.2
FOUNTAIN	5565	Mar	5
GUNNISON 1 N	7680	Mar	6.5
POWDERHORN	8094	Mar	5

AVG

[illegible]

Maximum with a window function!

A	B	C	D		F
Station	Elevation	Month	Snowfall		Max Elevation
BOULDER	5404	Jan	10.9		8094
DENVER WSFO AP	5325	Jan	7.3		8094
FOUNTAIN	5565	Jan	3.3		8094
GUNNISON 1 N	7680	Jan	12.5		8094
POWDERHORN	8094	Jan	7.2		8094
BOULDER	5404	Feb	11		8094
DENVER WSFO AP	5325	Feb	7	MAX →	8094
FOUNTAIN	5565	Feb	3		8094
GUNNISON 1 N	7680	Feb	9.5		8094
POWDERHORN	8094	Feb	6.9		8094
BOULDER	5404	Mar	17.8		8094
DENVER WSFO AP	5325	Mar	12.2		8094
FOUNTAIN	5565	Mar	5		8094
GUNNISON 1 N	7680	Mar	6.5		8094
POWDERHORN	8094	Mar	5		8094

Grouped aggregation

Grouped aggregation

- Often, we want to aggregate **one column** according to groupings defined by **a second column**
- Requires min of 3 ingredients
 - A grouping column
 - A data column
 - A summary method

Grouped aggregation

One grouping columns

Grouping column	Data column
A	
A	
A	
A	
B	
B	
B	
B	
C	
C	
C	
C	

Summary method



Group ID	Aggregation Outcome
A	Number 1
B	Number 2
C	Number 3

Grouped aggregation

Two grouping columns

Grouping column 1	Grouping column 2	Data column
A	Fast	
A	Fast	
A	Slow	
A	Slow	
B	Fast	
B	Fast	
B	Slow	
B	Slow	
C	Fast	
C	Fast	
C	Slow	
C	Slow	

Summary method



Group 1 ID	Group 2 ID	Aggregation Outcome
A	Fast	Number 1
A	Slow	Number 2
B	Fast	Number 3
B	Slow	Number 4
C	Fast	Number 5
C	Slow	Number 6

Grouped Aggregation

Avg Snowfall by Month

A	B	C	D
Station	Elevation	Month	Snowfall
BOULDER	5404	Jan	10.9
DENVER WSFO AP	5325	Jan	7.3
FOUNTAIN	5565	Jan	3.3
GUNNISON 1 N	7680	Jan	12.5
POWDERHORN	8094	Jan	7.2
BOULDER	5404	Feb	11
DENVER WSFO AP	5325	Feb	7
FOUNTAIN	5565	Feb	3
GUNNISON 1 N	7680	Feb	9.5
POWDERHORN	8094	Feb	6.9
BOULDER	5404	Mar	17.8
DENVER WSFO AP	5325	Mar	12.2
FOUNTAIN	5565	Mar	5
GUNNISON 1 N	7680	Mar	6.5
POWDERHORN	8094	Mar	5

AVG

G	H
Month	Avg Snowfall
Jan	8.24
Feb	7.48
Mar	9.3

Partitioned window functions

Partitioned window functions

- **Partitioned window function $\leftrightarrow$ Grouped Aggregation.** Again, we want to aggregate **one column** according to groupings (partitions) defined by **a second column**
- Requires min of 3 ingredients
 - A partition column
 - A data column
 - A summary method

Partitioned window function

One partition column

Grouping column	Data column
A	
A	
A	
A	
B	
B	
B	
B	
C	
C	
C	
C	

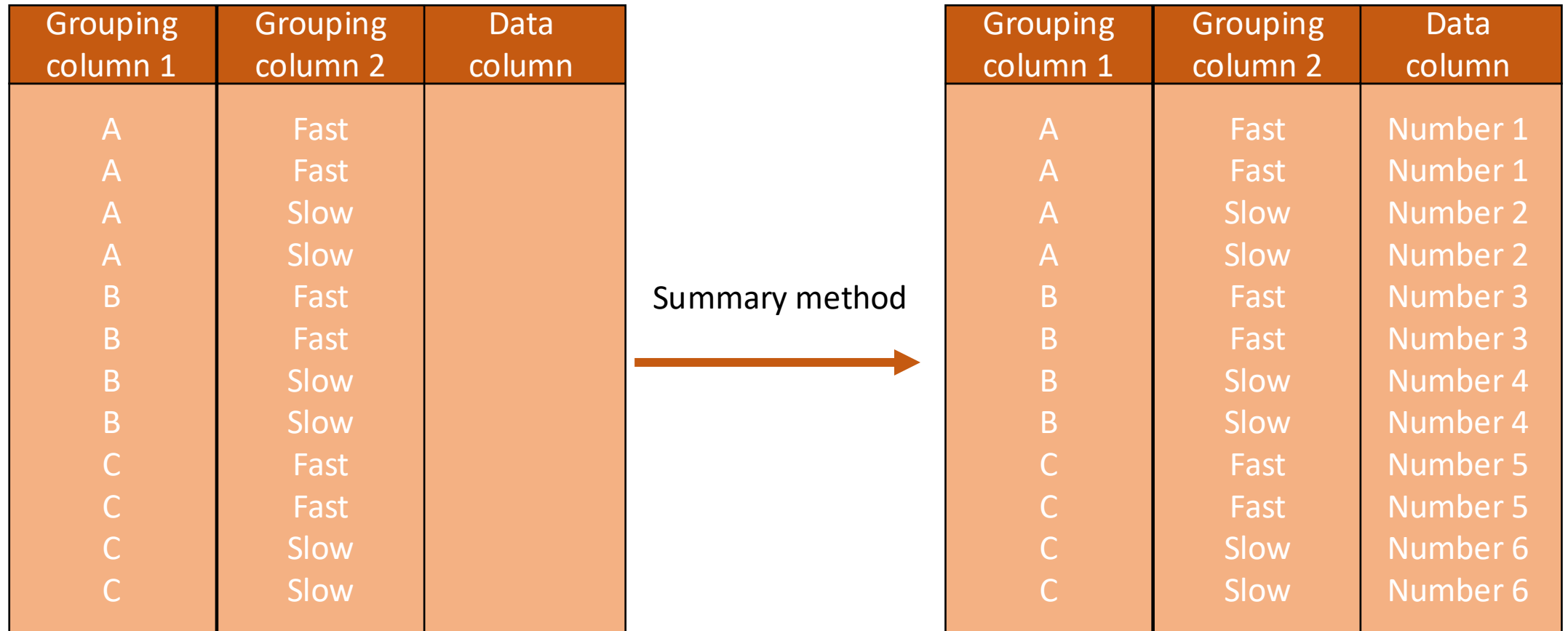
Summary method



Grouping column	Window outcome
A	Number 1
A	Number 1
A	Number 1
A	Number 1
B	Number 2
B	Number 2
B	Number 2
B	Number 2
C	Number 3
C	Number 3
C	Number 3
C	Number 3

Partitioned window function

Two partition column



Averaging with a window function!

Partition by Month

A	B	C	D
Station	Elevation	Month	Snowfall
BOULDER	5404	Jan	10.9
DENVER WSFO AP	5325	Jan	7.3
FOUNTAIN	5565	Jan	3.3
GUNNISON 1 N	7680	Jan	12.5
POWDERHORN	8094	Jan	7.2
BOULDER	5404	Feb	11
DENVER WSFO AP	5325	Feb	7
FOUNTAIN	5565	Feb	3
GUNNISON 1 N	7680	Feb	9.5
POWDERHORN	8094	Feb	6.9
BOULDER	5404	Mar	17.8
DENVER WSFO AP	5325	Mar	12.2
FOUNTAIN	5565	Mar	5
GUNNISON 1 N	7680	Mar	6.5
POWDERHORN	8094	Mar	5

AVG

E
Avg Snowfall
8.24
8.24
8.24
8.24
8.24
7.48
7.48
7.48
7.48
7.48
9.3
9.3
9.3
9.3
9.3

Pivot table

- **Pivot table $\leftrightarrow$ Grouped Aggregation.** Again, we want to
 - aggregate **one column**
 - according to groupings defined by a **second *pivot* column**
 - And optionally additional **grouping columns**
- Requires min of 3-4 ingredients
 - A pivot column
 - A data column
 - A summary method
 - 1 or more additional grouping columns

Pivot table

No grouping columns

Pivot column	Data column
Fast	
Fast	
Slow	
Slow	
Fast	
Fast	
Slow	
Slow	
Fast	
Fast	
Slow	
Slow	

Summary method



Pivot ID	
Fast	Slow
Number 1	Number 2

Pivot table

One grouping columns

Grouping column 1	Pivot column	Data column
A	Fast	
A	Fast	
A	Slow	
A	Slow	
B	Fast	
B	Fast	
B	Slow	
B	Slow	
C	Fast	
C	Fast	
C	Slow	
C	Slow	

Summary method



Group 1 ID	Pivot ID	
	Fast	Slow
A	Number 1	Number 2
B	Number 3	Number 4
C	Number 5	Number 6

Averaging with a pivot table!

Pivot on Month

A	B	C	D
Station	Elevation	Month	Snowfall
BOULDER	5404	Jan	10.9
DENVER WSFO AP	5325	Jan	7.3
FOUNTAIN	5565	Jan	3.3
GUNNISON 1 N	7680	Jan	12.5
POWDERHORN	8094	Jan	7.2
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DENVER WSFO AP	5325	Mar	12.2
FOUNTAIN	5565	Mar	5
GUNNISON 1 N	7680	Mar	6.5
POWDERHORN	8094	Mar	5

AVG

G	H	I
Jan	Feb	Mar
8.24	7.48	9.3